



Capstone Project 1

CMU-SE 450

Database Design

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Online Clinic Booking System

Submitted by

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A handwritten signature in black ink, appearing to be 'Thuan', written over a horizontal line.

Project Information

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V1.0	04/9/2021	Create Database Design
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TABLE OF CONTENT

Project Information.....	1
Document Infomation.....	2
1. Introduction	4
1.1. Purpose.....	4
1.2. Goal.....	4
1.3. Scope.....	4
1.4. Data storage platforms	4
1.5. Definition, Acronyms and Abbreviations.....	5
2. Database Design	5
2.1. Table Overview	5
2.2. Entity Relationship Diagram.....	0
2.3. Table Relationship Diagram	7
2.4. Detail.....	8
2.4.1. NguoiDung	8
2.4.2. BenhNhan	8
2.4.3. BacSi.....	8
2.4.4. PhongKham	9
2.4.5. LichKham	9
2.4.6. LichSu.....	9
2.4.7. DonThuoc	10
2.4.8. Benh.....	10
2.4.9. DanhGia.....	10
2.4.10. ThongBao	10

1. Introduction

1.1. Purpose

Place information system's database design document describes structure of database and file structure of system. Database Design document will introduce all attribute the System that will help developer and tester base on this design to implement and test.

1.2. Goal

To create database tables most accurate.

1.3. Scope

This Database Design Document provides the basic for "OnClinic" Database design.

It describes both logical and physical definition, non-functional issues, and the database interfaces; storage aspects are defined in the physical database design sections.

The tables performance considerations requirements. The following topics are covered in this document:

- Assumptions and decisions on database design.
- Entity-mapping.
- Table column definitions.
- Primary, unique and foreign key definitions.
- Column and row level validation rules (check constraints).
- Rule for populating specific columns (sequences, derivations, demoralized (column)).
- Interfaces and dependencies with other components.
- Data access description.

The database design is composed of definitions for database objects derived by mapping entities to tables attributes to columns, unique identifiers to unique keys and relationships to foreign keys.

During design, these initial definitions are enhanced to support the functionality described in the functional specification/ user stories and defined in the primary and supporting modules of the application high level design.

1.4. Data storage platforms

- Data of application is stored in PostgreSQL database
- In PostgreSQL, data structure is stored as tables that the database objects that behave as containers for the data, in which the data will be logically organized in rows and columns format.

1.5. Definition, Acronyms and Abbreviations

Abbreviations	Description	Comment
PK/FK	Primary/ Foreign Key	Use to indicate a file is a Primary or Foreign key in a table
ERD	Entity Relationship Diagram	Show the relationship between entities in the system

2. Database Design

2.1. Table Overview

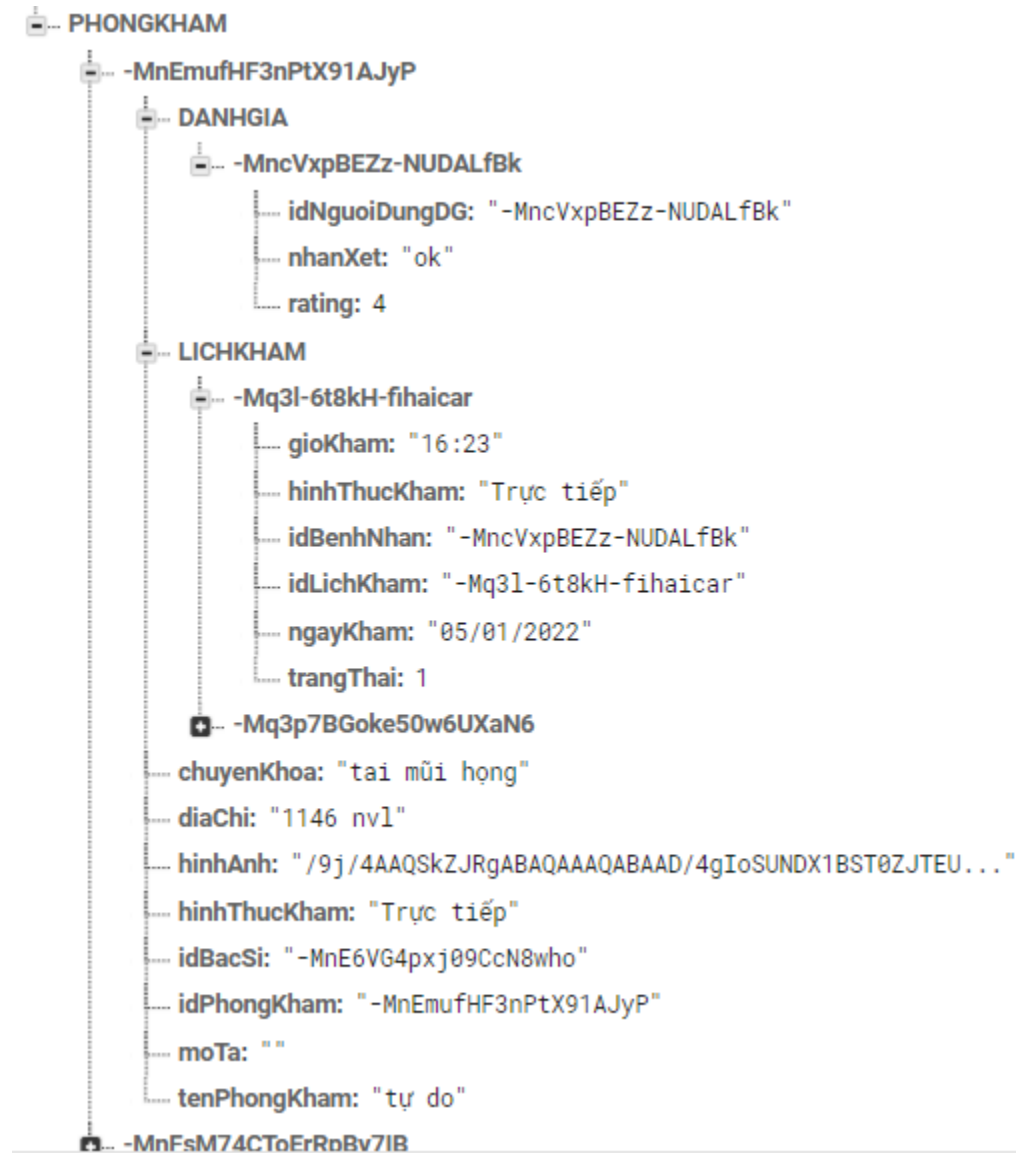
No.	Table name	Short Description
1	NGUOIDUNG	This table shows all information of User
2	PHONGKHAM	This table shows all information of Clinic
3	LICHKHAM	This table shows all information of Examination Schedule
4	DONTHUOC	This table shows all information of Medicine
5	LICHSU	This table shows all information of Medical History
6	BENH	This table shows all information of Disease
7	DANHGIA	This table shows all information of Vote and Evaluate

2.2. JSON Tree Structure

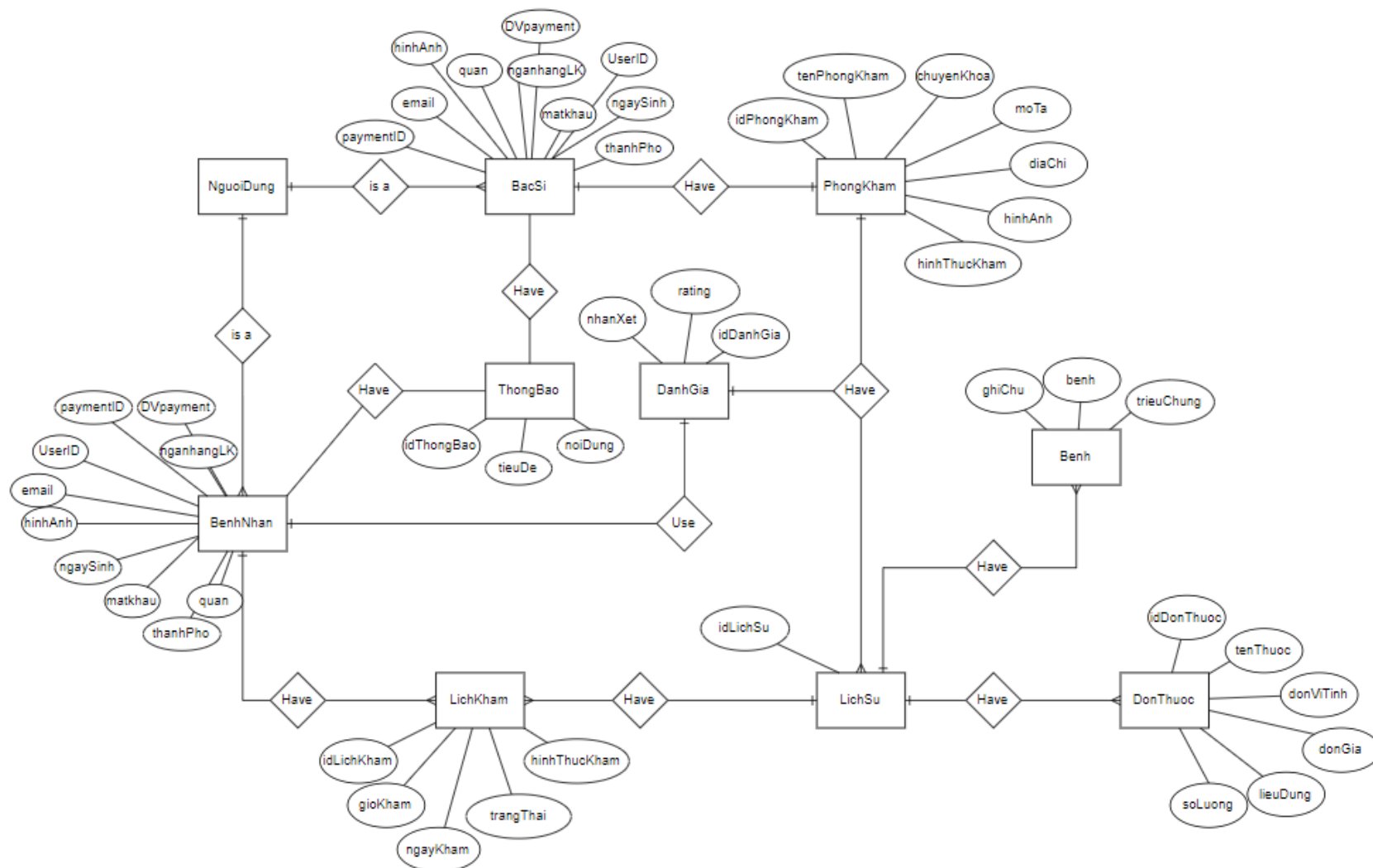
onclinic-180ee-default-rtdb



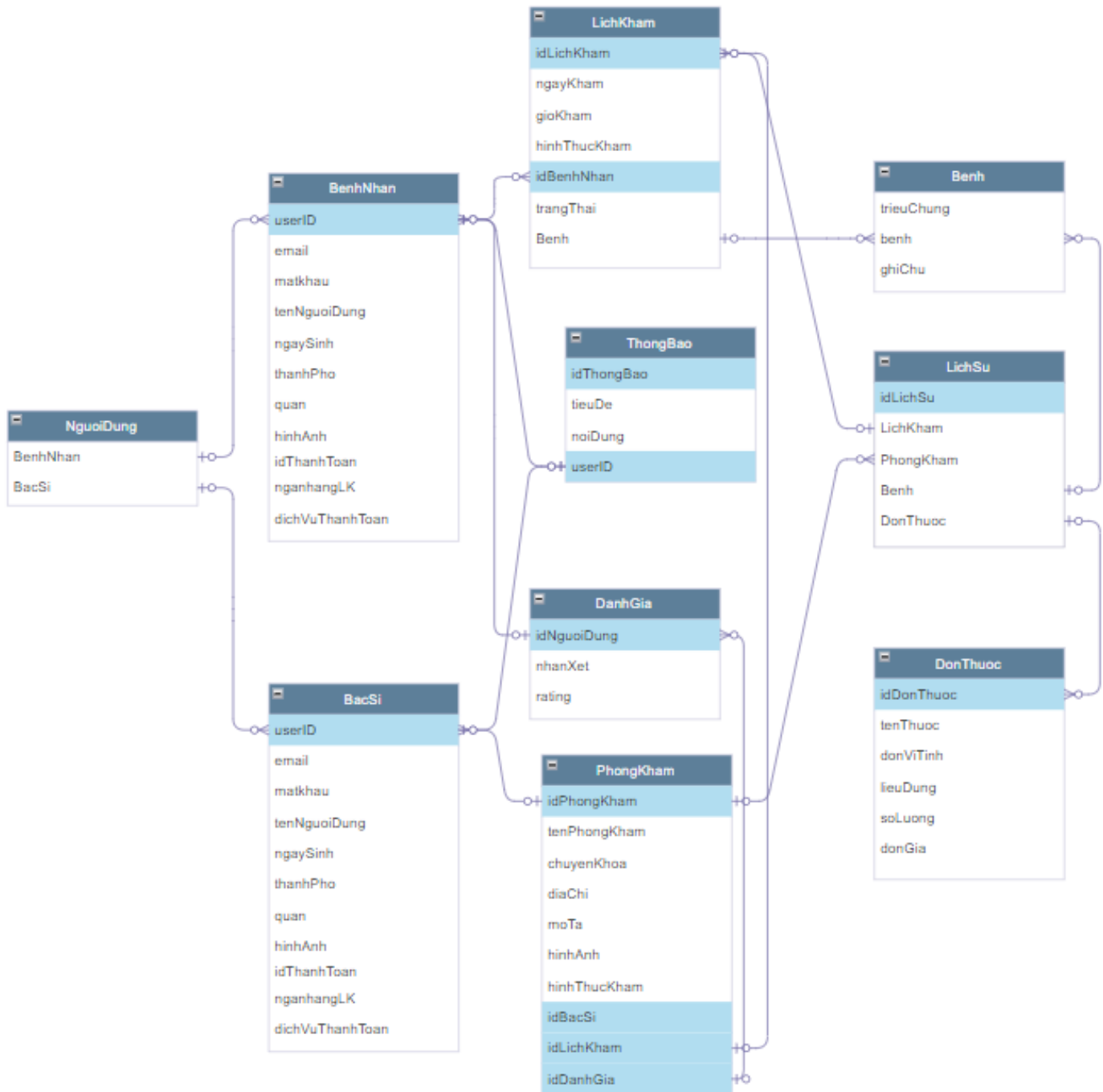




2.2. Entity Relationship Diagram



2.3. Table Relationship Diagram



2.4. Detail

2.4.1. NguoiDung

Attributes	Datatype	Null	Key	Default	Extra
BenhNhan	object				
BacSi	object				

2.4.2. BenhNhan

Attributes	Datatype	Null	Key	Default	Extra
userID	String	Not	PK, FK		
email	String	Not			
matkhau	String	Not			
tenNguoiDung	String	Not			
ngaySinh	String	Not			
thanhPho	String	Not			
quan	String	Not			
hinhAnh	String	Not			
idThanhToan	String				
nganhangLK	String				
dichVuThanhToan	String				

2.4.3. BacSi

Attributes	Datatype	Null	Key	Default	Extra
userID	String	Not	PK, FK		
email	String	Not			
matkhau	String	Not			
tenNguoiDung	String	Not			
ngaySinh	String	Not			
thanhPho	String	Not			
quan	String	Not			
hinhAnh	String	Not			
idThanhToan	String				
nganhangLK	String				
dichVuThanhToan	String				

2.4.4. PhongKham

Attributes	Datatype	Null	Key	Default	Extra
idPhongKham	String	Not	PK		
tenPhongKham	String	Not			
chuyenKhoa	String	Not			
diaChi	String	Not			
moTa	String				
hinhAnh	String	Not			
hinhThucKham	String	Not			
idBacSi	String	Not	FK		
idLichKham	String	Not	FK		
idDanhGia	String	Not	FK		

2.4.5. LichKham

Attributes	Datatype	Null	Key	Default	Extra
idLichKham	String	Not	PK		
ngayKham	String	Not			
gioKham	String	Not			
hinhThucKham	String	Not			
idBenhNhan	String	Not	FK		
trangThai	String	Not			
Benh	String	Not			

2.4.6. LichSu

Attributes	Datatype	Null	Key	Default	Extra
idLichSu	String	Not	PK		
LichKham	object	Not	FK		
PhongKham	object	Not	FK		
Benh	object				
DonThuoc	object				

2.4.7. DonThuoc

Attributes	Datatype	Null	Key	Default	Extra
idDonThuoc	String	Not	PK		
tenThuoc	String	Not			
donViTinh	String	Not			
lieuDung	String	Not			
soLuong	String	Not			
donGia	String	Not			

2.4.8. Benh

Attributes	Datatype	Null	Key	Default	Extra
trieuChung	String	Not			
benh	String	Not			
ghiChu	String				

2.4.9. DanhGia

Attributes	Datatype	Null	Key	Default	Extra
idNguoiDung	String	Not	PK		
nhânXet	String	Not			
rating	String	Not			

2.4.10. ThongBao

Attributes	Datatype	Null	Key	Default	Extra
idThongBao	String	Not	PK		
tieuDe	String	Not			
noiDung	String	Not			
userID	String	Not	FK		